

MIND MAPPING MATHEMATICS

SEM 1, 2017/2018

ALWAYS BE POSITIVE, SO LET'S WRITE YOUR NAME IN MODULUS; $|NAME|$

NAME :

TUTORIAL :

MY TARGET FOR PSPM 1 :

WITH HOPE,

Department of Mathematics, KMM

The HARDER you work, the LUCKIER you get!

Number System

$\frac{2}{7}, \frac{9}{5}$ and $8 = \frac{8}{1}$
 $= \{ \frac{p}{q}, p, q \in \mathbb{Z} \text{ and } q \neq 0 \}$

π, e and $\sqrt{3}$

$N = \{1, 2, 3, \dots\}$

$= \{2, 3, 5, 7, \dots\}$

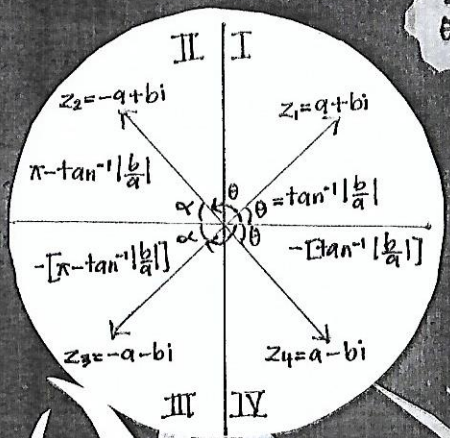
$\sqrt{p} \cdot \sqrt{q} = \sqrt{pq}$
 $\frac{\sqrt{p}}{\sqrt{q}} = \sqrt{\frac{p}{q}}$
 $p\sqrt{r} \pm q\sqrt{r} = (p \pm q)\sqrt{r}$

$x^m \cdot x^n = x^{m+n}$
 $x^m \div x^n = x^{m-n}$
 $(x^m)^n = x^{mn}$
 $(xy)^n = x^n \cdot y^n$
 $x^0 = 1$
 $x^{\frac{m}{n}} = \sqrt[n]{x^m} = (\sqrt[n]{x})^m$
 $x^{\frac{1}{n}} = \sqrt[n]{x}$
 $x^{-n} = \frac{1}{x^n}$
 $(\frac{x}{y})^n = \frac{x^n}{y^n}$

$\log_a xy = \log_a x + \log_a y$
 $\log_a \frac{x}{y} = \log_a x - \log_a y$
 $\log_a x^b = b \log_a x$
 $\log_a a = 1$
 $\log_a 1 = 0$
 $a^{\log_a x} = x$
 $\log_a x = \frac{\log_b x}{\log_b a}$

$W = \{0, 1, 2, 3, \dots\}$

$Z = \{\dots, -2, -1, 0, 1, 2, \dots\}$



$z_4 = a-bi$
 $\theta = -[\tan^{-1} \frac{b}{a}]$

$z_3 = -a-bi$
 $\theta = -[\pi - \tan^{-1} \frac{b}{a}]$

$z_2 = -a+bi$
 $\theta = \pi - \tan^{-1} \frac{b}{a}$

$z_1 = a+bi$
 $\theta = \tan^{-1} \frac{b}{a}$

INDICES
SURD AND
LOGARITHM.

LAW OF
INDICES

LAW OF
LOGARITHM

REAL
NO LINE

INTERVAL

$\cup$, UNION
 $\cap$, INTERSECTION

SET
NOTATION

$\{x = 2 < x < 4\}$

INTERVAL
NOTATION

(a, b)
 open interval

$[a, b]$
 close interval

$[a, b)$
 half-open interval

SURDS

→ Squaring both sides

ex: Solve $\sqrt{x+1} + 2 = \sqrt{x+3}$
 $(\sqrt{x+1} + 2)^2 = (\sqrt{x+3})^2$

→ Condition for surds
 If $f(x) = \sqrt{x}$, $x \geq 0$

Log / Ln

- i - Rearrange & Compare
- ii - Change to quadratic form

ex: Solve $2\log x = \log 4$ - i
 solve $(\log x)^2 + 3\log x = 4$ - ii
 let $\log x = a$

INDICES

- i - Rearrange & Compare (same base)
- ii - Rearrange & taking log (diff base)
- iii - Change to quadratic

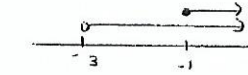
Solve $3^{x+2} = 9$ - i
 solve $3^{x+2} = 5$ - ii
 solve $(3^x)^2 + 3(3^x) = 4$
 let $3^x = a$ - iii

RATIO

ex: solve $\frac{x+1}{x+3} \geq 0$

NUMBER LINE

$x+1 \geq 0$ $x+3 > 0$
 $x \geq -1$ $x > -3$



$\{x: x < -3 \text{ or } x \geq -1\}$

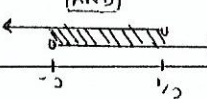
LINEAR

ex: solve $3x+4 \geq 2x-5$
 rearrange

solve $-4 < 3x < 5$
 rearrange $-\frac{4}{3} < x < \frac{5}{3}$

solve $1 < -2x < 4$
 number line

$\{x: -2 < x < \frac{1}{2}\}$



ASOIF MATHEMATICS CHP 2

EQUATIONS, INEQUALITIES & ABSOLUTE VALUES

ABSOLUTE Inequalities

- 1) $|ax+b| > cx+d$
- 2) $|ax+b| > |cx+d|$
- 3) $|ax^2+bx+c| > d$

$|x| \geq a \Rightarrow x \geq a \text{ or } x \leq -a$
 $|x| \leq a \Rightarrow x \leq a \text{ and } x \geq -a$

ex: $|4x-1| \geq 3$

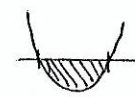
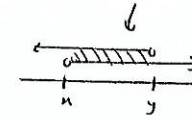
$4x-3 \geq 3$ or $4x-3 \leq -3$

Definition

$|a| = \sqrt{a^2}$

- Use when both sides have $| |$
- Use when question from quadratic equation → $| |$

QUADRATIC



	$(-\infty, -1]$	$(-1, 1)$	$[1, \infty)$
$ax+b$	+	+	-
$c-x$	-	+	-
	-	+	-

PROPERTIES of ABSOLUTE VALUE

- 1) $|a| \geq 0$
 ex: $|7| = 7$
- 2) $|a| = |-a|$
 ex: $|3| = |-3|$
 $3 = 3$
- 3) $|a+b| = |b+a|$
 ex: $|7+2| = |2+7|$
 $9 = 9$

ABSOLUTE Equations

- 1) $|ax+b| = c$
- 2) $|ax+b| = |cx+d|$
- 3) $|ax^2+bx+c| = d$

$|x| = a \Rightarrow x = a, x = -a$

ex: $|4x-3| = 9$

$4x-3 = 9$ $4x-3 = -9$
 $x = 3$ $x = -3/2$

Arithmetic

SEQUENCES

Geometric

Common difference,

$$d = T_n - T_{n-1}$$

eg. $d = T_2 - T_1$

Common ratio,

$$r = \frac{T_n}{T_{n-1}}$$

eg. $r = \frac{T_2}{T_1}$

Say $T_1 = a$, with a as the first term,

$\therefore n^{\text{th}}$ term, $T_n = a + (n - 1)d$

$\therefore n^{\text{th}}$ term, $T_n = ar^{n-1}$

Arithmetic

SERIES
is the sum of sequence

Geometric

Sum of n terms,

$$S_n = \frac{n}{2} [2a + (n - 1)d]$$

$$S_n = \frac{n}{2} (a + l), l \text{ is the last term}$$

Sum of n terms,

$$S_n = \frac{a(1 - r^n)}{1 - r}, r < 1$$

$$S_n = \frac{a(r^n - 1)}{r - 1}, r > 1$$

Known difference of consecutive
arithmetic S_n

defined the value of term n ,

$$T_n = S_n - S_{n-1}$$

Sum to infinity of geometric series,

$$S_\infty = \frac{a}{1 - r}, \text{ if } -1 < r < 1$$

Applications

Binomial expansion,

$$T_{r+1} = \binom{n}{r} p^{n-r} \cdot q^r, \text{ if } (p + q)^n$$

if n is positive integer,
 $n \in \mathbb{N}$

Binomial

if n is not a positive integer,
 $n \in \mathbb{Z}^-, n \in \mathbb{Q}$

$$C_{ij} = \begin{bmatrix} +M_{11} & -M_{12} & +M_{13} \\ -M_{21} & +M_{22} & -M_{23} \\ +M_{31} & -M_{32} & +M_{33} \end{bmatrix}$$

NB: ∇ upper triangle
 Δ lower triangle
 NB: diagonal

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$|A| = ad - bc$$

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$i = \text{row}, j = \text{column}$

$$m_{ij} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$|A| = \sum_{j=1}^n a_{ij} C_{ij}$$

* choose any row/column
 ex: 1st column

$$\therefore |A| = a_{11}m_{11} - a_{12}m_{12} + a_{13}m_{13}$$

Row $\leftrightarrow$ Column

- $(A^T)^T = A$
- $(A+B)^T = A^T + B^T$
- $(kA)^T = kA^T$, k is a scalar
- $(AB)^T = B^T A^T$

$$[A|I] \xrightarrow{\text{ERO}} [I|A^{-1}]$$

* Steps (1-9)

$$\left[\begin{array}{ccc|ccc} 7 & 4 & 5 & 1 & 0 & 0 \\ 1 & 8 & 6 & 0 & 1 & 0 \\ 2 & 3 & 9 & 0 & 0 & 1 \end{array} \right]$$

$$1 \leftrightarrow 2, 3 \leftrightarrow 4, 5 \leftrightarrow 6$$

$$7 \leftrightarrow 8, 8 \leftrightarrow 9$$

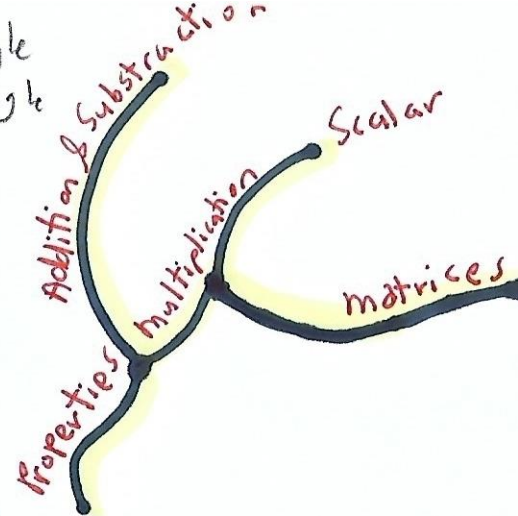
$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{Adj}(A)$$

$$* \text{Adj}(A) = [C_{ij}]^T$$

Transpose matrix

Matrices & Systems of Linear Equations



- * $A_{m \times n} \times B_{n \times p} = AB_{m \times p}$
- $AB = 0$, if A zero matrix
- $AB \neq BA$
- $A I = I A = A$
- $A^2 = AA$
- $A^p A^q = A^{p+q}$
- $(A^p)^q = A^{pq}$

Unique Solution
 if $|A| \neq 0$
 $\therefore$ shown



$$\text{From } AX = B$$

Inverse (multiply with A^{-1})

$$A^{-1}AX = A^{-1}B$$

$$IX = A^{-1}B$$

$$\therefore X = A^{-1}B$$

$$A^{-1} = ? \text{ (adj/ERO/deduce)}$$

NB: Solving systems of linear equation

Gauss Jordan Elimination

$$[A|B] \xrightarrow{\text{GJE}} [I|X]$$

$$\text{From } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\therefore x = ?, y = ?, z = ?$$

A^{-1} exist if $|A| \neq 0$
 (A: non singular matrix)

ex: deduce A^{-1} using properties

$$A^2 - 4A + 3I = 0 \leftarrow \text{matrix equation}$$

$$\underbrace{A^{-1}A} I A - 4 \underbrace{A^{-1}A} I + 3 \underbrace{A^{-1}I} A^{-1} = 0$$

$$IA - 4I + 3A^{-1} = 0$$

$$A - 4I + 3A^{-1} = 0$$

$$\therefore A^{-1} = \frac{1}{3} (4I - A)$$

EXIST!

Determining a function

- Vertical Line Test
(Succeed if cuts at 1 point ONLY)

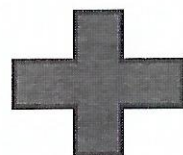
Determining 1 – 1 function

- Horizontal Line Test
(Succeed if cuts at 1 point ONLY)
- Algebraic Approach
(LET $f(x_1)=f(x_2)$
 $\Rightarrow$ if $x_1=x_2$, then $f(x)$ is 1 – 1 function.

FUNCTION

Inverse function

$$\rightarrow f(f^{-1}(x))=x$$



GRAPHS

LINEAR: $y = x$

QUADRATICS: $y = x^2$

Completing The Square is VITAL!

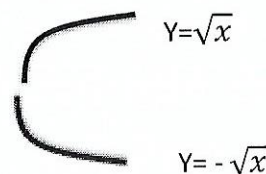
CUBIC: $y = x^3$

RECIPROCAL: $y = \frac{1}{x}$

ALWAYS find both horizontal and vertical asymptotes!

SURD: $y = \sqrt{x}$

surd graph is like quadratic graph, but only half of it and lying horizontally



LOGARITHM: $y = \ln x$

EXPONENTIAL: $y = e^x$

ABSOLUTE: $y = |x|$

*V shaped graph

Use Q I A method to sketch

Q – determine the quadrant of the starting point. This point will be far from axes.

I – determine the x & y intercepts.

A – determine the horizontal / vertical asymptote

Combining these functions creates a piecewise function.

- Division

$$\begin{array}{r} Q(x) \\ D(x) \overline{) P(x)} \\ \underline{R(x)} \end{array}$$

$$\frac{P(x)}{D(x)} = Q(x) + \frac{R(x)}{D(x)}$$

$$P(x) = D(x)Q(x) + R(x)$$

- Addition
- Subtraction
- Multiplication

6.1 Polynomials

POLYNOMIALS

6.2 Remainder Theorem, Factor Theorem and Zeroes of Polynomial

Remainder Theorem

If a polynomial $P(x)$ is divided by a linear factor $x - a$, then the remainder is $P(a) \Rightarrow P(a) = R(x)$

Factor Theorem

If a polynomial $P(x)$ is exactly divisible by $x - a$, then the remainder is 0. $\Rightarrow P(a) = 0$

Zeroes of Polynomial

If $P(x) = (x - a)(x - b)(x - c) = 0$,

- Factors : $x - a, x - b, x - c$
- Roots : $x = a, x = b, x = c$
- Zeroes : a, b, c

6.3 Partial Fractions

Improper fractions

Long Division

Types of fractions

Proper fractions

Linear factor

Quadratic factor

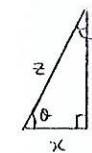
Repeated Linear factor
 $b^2 - 4ac = 0$

Can be factorized
 $b^2 - 4ac > 0$

Cannot be factorized
 $b^2 - 4ac < 0$

ZHJ @ KMM

Trigonometric ratios



$$a) \sin \theta = \frac{y}{z} \quad d) \operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{z}{y}$$

$$b) \cos \theta = \frac{x}{z} \quad e) \sec \theta = \frac{1}{\cos \theta} = \frac{z}{x}$$

$$c) \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{y}{x} \quad f) \cot \theta = \frac{1}{\tan \theta} = \frac{x}{y}$$

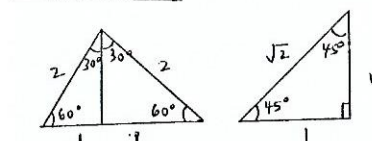
Relationship

$$a) \sin (90^\circ - \theta) = \cos \theta$$

$$b) \cos (90^\circ - \theta) = \sin \theta$$

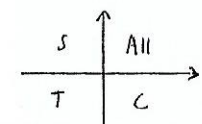
$$c) \tan (90^\circ - \theta) = \cot \theta$$

Special angles



	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞

Evaluate Trigonometric functions



$$a) \sin (-\theta) = -\sin \theta$$

$$b) \cos (-\theta) = \cos \theta$$

$$c) \tan (-\theta) = -\tan \theta$$

Pythagorean Identities

$$a) \sin^2 \theta + \cos^2 \theta = 1$$

$$b) \cot^2 \theta + 1 = \operatorname{cosec}^2 \theta$$

$$c) \tan^2 \theta + 1 = \sec^2 \theta$$

① Solve equations (Basic trigonometry)

$$a) \sin \theta = k$$

$$b) \cos \theta = k$$

$$c) \tan \theta = k$$

② Solve equations (quadratic form)

① Sum/Difference Formulae

$$a) \sin (A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$b) \cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$c) \tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

② Double Angle Formulae

$$a) \sin 2A = 2 \sin A \cos A$$

$$b) \cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$$

$$c) \tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

③ Half Angle Formulae

$$a) \sin \frac{A}{2} = 2 \sin \frac{A}{2} \cos \frac{A}{2}$$

$$b) \cos A = \cos^2 \frac{A}{2} - \sin^2 \frac{A}{2} = 1 - 2\sin^2 \frac{A}{2} = 2\cos^2 \frac{A}{2} - 1$$

$$c) \tan A = \frac{2 \tan \frac{A}{2}}{1 - \tan^2 \frac{A}{2}}$$

④ Factor Formulae

$$a) \sin A + \sin B = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$$

$$b) \sin A - \sin B = 2 \cos \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)$$

$$c) \cos A + \cos B = 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$$

$$d) \cos A - \cos B = -2 \sin \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)$$

⑤ Express/solve

$$a) a \sin \theta + b \cos \theta = R \sin (\theta + \alpha) \text{ or } R \cos (\theta - \alpha)$$

$$b) a \sin 2\theta + b \cos 2\theta = R \sin (2\theta + \alpha) \text{ or } R \cos (2\theta - \alpha)$$

$$* R = \sqrt{a^2 + b^2}$$

$$* \alpha = \tan^{-1} \left(\frac{b}{a} \right)$$

⑥ Min/max values

$$a) \text{ min value}$$

$$\sin \frac{3\pi}{2} = -1$$

$$\cos \pi = -1$$

$$b) \text{ max value}$$

$$\sin \frac{\pi}{2} = 1$$

$$\cos 0 = 1$$

$$\cos 2\pi = 1$$

A. Trigonometric Ratios and Identities

7. Trigonometric Functions

B. Compound Angle

C. Solutions of Trigonometric Functions

① Using t substitution

$$a) \text{ if } t = \tan \frac{\theta}{2}$$

* use half angle formula

$$\tan \theta = \frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}} = \frac{2t}{1-t^2}$$

$$b) \text{ if } t = \tan \theta$$

* use double angle formula

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{2t}{1-t^2}$$

Finding Limits

$$1) \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\text{constant}}{\text{constant}} = \text{constant}$$

$$2) \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{k} = 0$$

$$3) \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0} \Rightarrow \text{simplify / define factorize / conjugate}$$

$$4) \lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = \frac{k}{0} = +\infty \text{ or } -\infty$$

$\Rightarrow$ sub : nearest value of x

$$5) \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{\infty}{\infty} \Rightarrow \text{divide each terms by the highest degree of } x \text{ from the denom.}$$

ASYMPTOTES

$y=b$ is a HA
if $\lim_{x \rightarrow \pm\infty} f(x) = b$

[Horizontal]

Vertical

$x=a$ is a VA
if $\lim_{x \rightarrow a^+} f(x) = \pm\infty$
or
 $\lim_{x \rightarrow a^-} f(x) = \pm\infty$

Existence of a limit

$\lim_{x \rightarrow a} f(x)$ exist if

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$$

LIMIT & CONTINUITY

Chap. 8

Continuity

3 conditions

$f(a)$ is defined

$\lim_{x \rightarrow a} f(x)$ exist

$$\lim_{x \rightarrow a} f(x) = f(a)$$

PROPERTIES

$$\bullet \lim_{x \rightarrow a} c = c$$

$$\bullet \lim_{x \rightarrow a} x = a$$

$$\bullet \lim_{x \rightarrow a} c f(x) = c \lim_{x \rightarrow a} f(x)$$

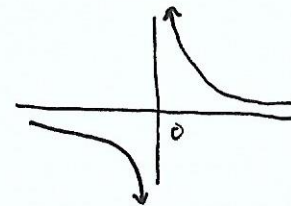
$$\bullet \lim_{x \rightarrow a} x^n = a^n$$

$$\bullet \lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$$

$$\bullet \lim_{x \rightarrow a} f(x)g(x) = \lim_{x \rightarrow a} f(x) \times \lim_{x \rightarrow a} g(x)$$

$$\bullet \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

Ex. of a graph $f(x) = \frac{1}{x}$



$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty \text{ (up)}$$

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty \text{ (down)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

First Principles

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Derivative of $f(x)$ at $x=a$

9.1

Topic 9: Differentiation

Rules of Differentiation

constant rule

sum rule

product rule

quotient rule

chain rule

power rule

- ① $f(x) = k$; $f'(x) = 0$, k - constant
- ② $f(x) = x^n$; $f'(x) = nx^{n-1}$, $n \in \mathbb{R}$
- ③ $f(x) = kx^n$; $f'(x) = nkx^{n-1}$
- ④ $f(x) = u+v$; $f'(x) = u' + v'$
- ⑤ $f(x) = uv$; $f'(x) = uv' + vu'$
- ⑥ $f(x) = \frac{u}{v}$; $f'(x) = \frac{vu' - uv'}{v^2}$
- ⑦ If $y = f(u)$; $u = g(x)$;
 $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
- ⑧ If $y = u^n$, u is a function of x
 $\frac{dy}{dx} = nu^{n-1} \left(\frac{du}{dx} \right)$ 4 STEPS

9.5

Parametric Differentiation

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \times \frac{dt}{dx}$$

9.2

Higher Order Derivative

$$\frac{d^2y}{dx^2}$$

- second order differentiation

$$\frac{d^3y}{dx^3}$$

- third order differentiation

9.4

Implicit Differentiation

Function not in the form $y = f(x)$	Function in the form $y = f(x)$
Implicit function eg: $y = x^2 + y^3$	Explicit function eg: $y = 3x + 1$

method

$$\frac{d}{dx}(y^n) = ny^{n-1} \left(\frac{dy}{dx} \right)$$

Derivative of trigonometric functions

$$\begin{aligned} \frac{d}{dx}(\sin x) &= \cos x \\ \frac{d}{dx}(\cos x) &= -\sin x \\ \frac{d}{dx}(\tan x) &= \sec^2 x \\ \frac{d}{dx}(\cot x) &= -\operatorname{cosec}^2 x \\ \frac{d}{dx}(\operatorname{cosec} x) &= -\operatorname{cosec} x \cot x \\ \frac{d}{dx}(\sec x) &= \sec x \tan x \end{aligned}$$

9.3

Derivative of log functions

$$\begin{aligned} \frac{d}{dx}(\ln x) &= \frac{1}{x} \\ \frac{d}{dx}[\ln(f(x))] &= \frac{f'(x)}{f(x)} \end{aligned}$$

Derivative of exponents

$$\begin{aligned} \frac{d}{dx}(a^x) &= a^x \ln a \\ \frac{d}{dx}(e^x) &= e^x \\ \frac{d}{dx}(e^{f(x)}) &= f'(x) e^x \end{aligned}$$

To find the **equation of a tangent or normal** to a curve:

- Find dy/dx
- Substitute the x coordinate of the required point.
- Use $y-y_1=m(x-x_1)$ to find the equation of the tangent.
- For the **normal** use $y-y_1=m'(x-x_1)$ with $m'=-1/m$

To find a **stationary point, turning point or maximum/minimum value** :

- Find dy/dx
- Set dy/dx equal to 0 and solve the resulting equation to find the x coordinates of any T.Ps
- Substitute the x coordinates into the equation of the curve to find the corresponding y coordinates.

DIFFERENTIATION
 $\frac{dy}{dx}$ or $f'(x)$

Gives the **gradient function of a curve.**

Substitute the x coordinate to find the gradient at a particular point on the curve

To find the values of x for which a **function is increasing or decreasing**:

- Find dy/dx
- Solve $dy/dx > 0$ to find the range of values of x for which the function is increasing
- Solve $dy/dx < 0$ to find the values of x for which the function is decreasing

To determine the **nature of a TP**:

- Find d^2y/dx^2 or $f''(x)$
- Substitute the x coordinate of the TP into $f''(x)$
- If $f''(x) > 0$ then TP is a Min
- If $f''(x) < 0$ then TP is a Max
- If $f''(x) = 0$ then you cannot tell and will need to consider the gradient before and after the TP